

HW 2: Margin of Error

Directions: For the following multiple-choice problems, show your work. You will not get full credit for only circling an answer.

_____ 1) A survey given out before the 2024 election asked Americans if they would be willing to pay higher taxes if the government offered free medical care for all citizens. Of the 2500 people polled, the survey produced a confidence interval of $0.35 \pm 2.5\%$. Which of the following is the closest to the confidence level this poll used?

- (A) 87.5%
- (B) 91%
- (C) 95%
- (D) 97%
- (E) 99%

_____ 2) The Environmental Protection Agency wants to conduct a survey to gauge public perception of its' mission and vision. From previous studies, they know that they need to poll at least 300 people to achieve their desired level of confidence. If they want to keep the same level of confidence but divide the margin of error by one fourth, how many people will they have to poll?

- (A) 75
- (B) 960
- (C) 1,200
- (D) 4,800
- (E) 9,600

_____ 3) The population of the United States is approximately 330 million people. A research firm wants to determine the opinions of all Americans on a new public policy, with 90% confidence, and a margin of error of 2%. Which of the following is the appropriate number of people the firm should survey?

- (A) 3,300,000
- (B) 1692
- (C) 1,692,000
- (D) 3,300
- (E) 133

_____ 4) A newspaper is interested in the unemployment rate in your state. Your local newspaper reports that in an SRS of 400 people in your city, they have a 95% confidence interval of $8\% \pm 3\%$. Which of the following is a true statement regarding the margin of error?

- (A) There is a 95% probability that the true unemployment rate is between 5% and 11%.
- (B) If your newspaper used 300 people instead of 400 people, the margin of error will decrease.
- (C) The margin of error communicates how reliable your data collections methods are; a lower margin of error will indicate that your sampling methods are unbiased.
- (D) The margin of error accounts for natural sampling variability and does not rely on population data.
- (E) In repeated samplings of 400 people, we would expect 95% of the resulting confidence intervals to have a margin of error of 3%.

5) Net neutrality is a concept that all Internet traffic should be treated equally – namely, no Internet service provider (like Verizon, Comcast, etc.) should be able to decide how Internet traffic should move based on where it is coming from or where it is going to. In a large-scale online survey, 175,000 Internet users responded, and 96% of them agree in favor of net neutrality. The survey results gave a confidence interval with a margin of error of 0.15%.

(a) Is the number 96% a parameter or statistic? Explain

(b) Without doing any calculations, explain why the margin of error is so small.

(c) How confident should you be that the true proportion of internet users who favor net neutrality is between 95.85% and 96.15%?